

Auger and Electron Energy Loss Spectroscopy

Advanced Laboratory Course for Master Student, WS 2022

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1 Introduction

Auger effect was first observed by Pierre Auger in the mid-1920's. Auger electron spectroscopy is based on this effect. When a photon of sufficient energy is incident on an atom, then the core electron is removed leaving a vacancy. The electron from higher energy level may fall into the vacancy resulting in the release of characteristic X-Rays. Latter this characteristic X-Rays will help in the liberation of second electron thus producing Auger effect. It is used for checking the surface cleanliness and finding the surface composition of the material.

2 Auger Electron Spectroscopy

The main principle is depicted in figure1. Mainly in our experiment we focus on aluminum metal. The incident electron beam knocks out an electron from K-shell thus creating a hole. To fill the hole electron from next higher shell that is L-shell comes into the K-shell. The energy of characteristic X-radiation is released which is called as X-ray fluorescence. This energy is transmitted to second electron in a radiation less process which may leave the atom with a characteristic kinetic energy. This second electron is called an Auger electron. In quantum dynamics we can write Auger process in following way:

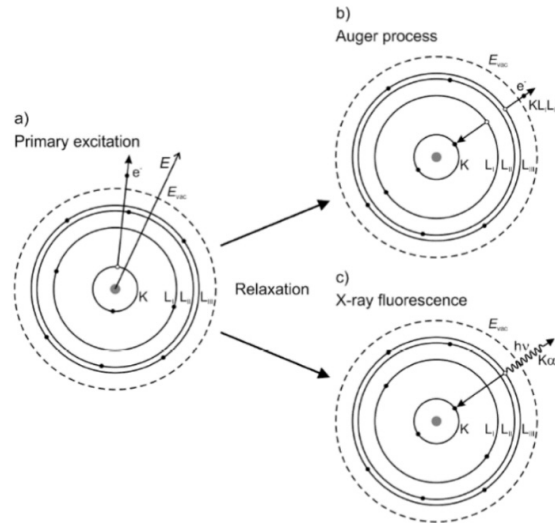


Figure 1: Phenomenon of AES process involving three shells KLL [5]

$$\langle \psi_1 s(r_1) \psi_2 a(r_1) | \hat{A} | \psi_2 b(r_2) e l(r_2) \rangle$$

where $\psi_{1s}(r_1)$ as the wavefunction of the K-shell hole $\psi_{2a}(r_1)$ is the wavefunction of the electron filling the K-shell hole $\hat{A}$ is an operator $\psi_{2b}(r_2)$ the second hole wave function $\epsilon l(r_2)$ as the emitted electron.

2.1 Surface Sensitivity

A surface sensitive technique is more sensitive to atoms which are located on the surface than to the bulk in which atoms are far away from the surface. The average distance travelled by an electron through a solid before it is inelastically scattered is called Inelastic Mean Free Path (IMFP). It depends on initial kinetic energy of the electron and the nature of the materials. The probability of the electron travelling a distance, d , through the solid without undergoing scattering is governed by the following equation:

$$P(d) = e^{\frac{-d}{\lambda(E)}} \quad (1)$$

Where λ is the IMFP for the electron energy E , (here: $\lambda = f(E)$). This IMFP is only related to the movement of electrons in solid. The log-log plot clearly shows how the IMFP varies at low kinetic energies. The probability of inelastic scattering decreases at lower energies since the electron has insufficient energy to cause plasmon excitation which is the main scattering mechanism in metals as a consequence the distance between inelastic collisions and the IMFP increases. In metals the IMFP is less than 10 Angstroms for electron energies in the range $15 < E/\text{eV} < 350$. This means experiments involving generation and detection of electrons of such energies will be surface sensitive. Since AES is an electron core-level spectroscopy, we use 5keV electrons for excitation which we can obtained from photons, electrons or even by ion bombardment. From the IMFP of electrons, the observation of Auger electrons with a kinetic energy around 1000 eV means an observation range of about 10 atomic layers below the surface. So probing depths in case of AES are in the range of 10-30 Angstrom

2.2 Nomenclature for Auger transitions

Here we are describing in terms of Aluminum which has KLM shell whereas L has two $L_1, L_{2,3}$ and M has M_1 and M_2 . We generally denote the transitions as KLL where the first letter comes from the shell of the created hole, the second letter comes from the shell

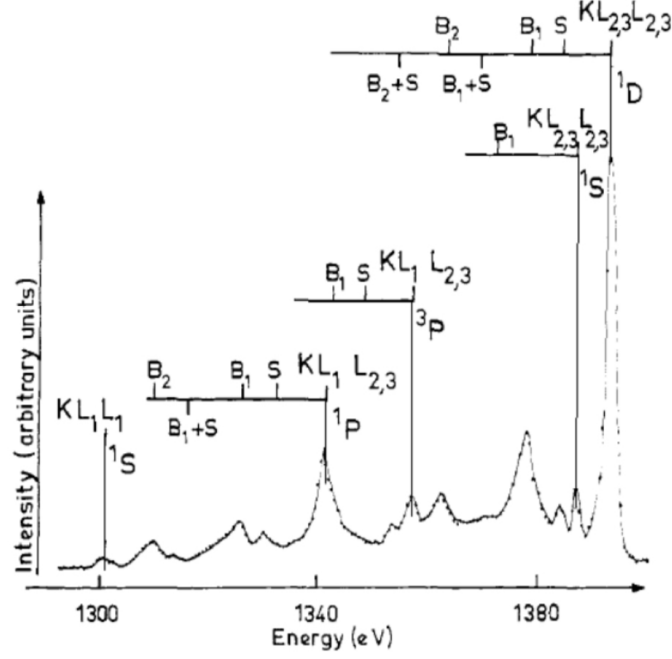


Figure 2: KLL Auger spectra of Al [5]

of the relaxing electron and the third letter comes from the shell of the emitted electron. The kinetic energy of the Auger-electron depends only on the participating shells and not on the order of the second and third letter of the participating shell.

$$E_{Auger} = E_{hole} - E_{Binding1} - E_{binding2}$$

Auger transitions of Aluminum is given in a table below:

Ground State	Final State	Electron	Allowed	Name
$1s^1 2s^2 p^6 {}^2S_{1/2}^e$	$1s^2 2s^0 2p^6 {}^1S_0^e$	$\epsilon s_{1/2}^e$	Yes	$KL_1 L_1 ({}^1S)$
$1s^1 2s^2 p^6 {}^2S_{1/2}^e$	$1s^2 2s^1 2p^5 {}^1P_1^0$	$\epsilon p_{3/2;1/2}^0$	Yes	$KL_1 L_{2,3} ({}^1P)$
$1s^1 2s^2 p^6 {}^2S_{1/2}^e$	$1s^2 2s^1 2p^5 {}^3P_{2;1;0}^0$	$\epsilon p_{3/2;1/2}^0$	Yes	$KL_1 L_{2,3} ({}^3P)$
$1s^1 2s^2 p^6 {}^2S_{1/2}^e$	$1s^2 2s^2 p^4 {}^1S_0^e$	$\epsilon s_{1/2}^e$	Yes	$KL_{2,3} L_{2,3} ({}^1S)$
$1s^1 2s^2 p^6 {}^2S_{1/2}^e$	$1s^2 2s^2 p^4 {}^1D_2^e$	$\epsilon d_{5/2;3/2}^e$	Yes	$KL_{2,3} L_{2,3} ({}^1D)$
$1s^1 2s^2 p^6 {}^2S_{1/2}^e$	$1s^2 2s^2 2p^4 {}^3P_{2;1;0}^0$	$\epsilon p_{3/2;1/2}^e$	No	$KL_{2,3} L_{2,3} ({}^3P)$

3 Electron Energy Loss Spectroscopy

James Hillier and RF Baker invented this technique in 1940. This method uses to study the inelastic scattering of fast electrons after surface excitation and scattering from ad-

sorbed layers. The Scattered electron may be deflected with smaller angles which then can be read through electron spectrometer. This finally gives us the reasons about the energy loss through this scattering event. This method of spectroscopy is mainly used for the atoms ranging from carbon to d-block elements which includes Aluminium.

3.1 Plasmons

Plasmons are quasi-particles coming into existence when an electron passes through the delocalized electrons in metals known as electron gas and gave rise to the oscillations. The auger electrons while passing through this delocalized electron gas can also lead to the formation of plasmons. There are two types of plasmons called bulk plasmons which appear deep in the material and surface plasmons which are created between the interface of the metal and dielectric material. It implies that the energy needed to excite the bulk plasmons is greater than to excite the surface plasmons. The approximate values for aluminium is 15.2eV and 10.4eV for bulk plasmon and surface plasmon respectively. The plasmon can be described with the harmonic equation of oscillation as

$$E = h\omega_p = h\sqrt{\frac{ne^2}{m_e\epsilon}} \quad (2)$$

where, h planks constant, ω_p is the plasmon frequency, n density of electrons, e the elementary charge on electron[1]

3.2 L-S Coupling

The coupling plays a main role here representing interaction of electron motion with its spin. Coupling determines the splitting of energy levels under the application of magnetic field. For the case of KLL transitions, it can exhibit the various types of coupling, but for the atoms having atomic numbers less than 30 we mostly restrict ourselves to L-S coupling, where L defines the total angular momentum coming from every addition of single angular momentum interactions of every electron that is 'l'. And same can be calculated for total spin angular momentum S by calculating it from spin angular associated with every electron that is 's'. Finally through L-S coupling we get the total associated angular momentum with atom that is J , where $J = L + S$. There are five peaks expected in this experiment for KLL process

- $KL_{2;3}L_{2;3}$
- $KL_1L_{2;3}$
- $KL_1L_{2;3}$.
- $KL_1L_{2;3}$
- KL_1L_1

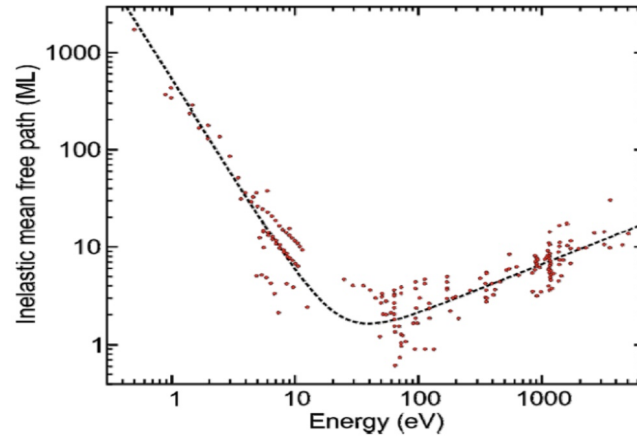


Figure 3: Universal IMFP curve with energy to detect the surface properties [5]

3.3 Inelastic Electron Free Path

The electron made incident on the metal surface will lose energy on interaction by different means like plasmon excitation, electron-hole pair formation and vibrational excitation. Depending upon the depth of the metal the incident electron reaches, the intensity of electrons decreases in a way of exponential damping:

$$I(d) = I_0 e^{\frac{-d}{\lambda(E)}} \quad (3)$$

where d is the distance electron reaches into the metal and $\lambda(E)$ is inelastic electron mean free path states the distance at which the intensity of the beam becomes $\frac{1}{e}$ of its initial value.

4 Experimental Setup

The experimental setup consists of following apparatus

4.1 Ultra-High Vacuum Chamber(UHV)

The heart of this experiment is ultra high vacuum chamber which is used to conduct auger electron spectroscopy to attain a clean surface. A low pressure around 10^{-10} (mbar) is maintained and all the equipment are placed in it. The main parts of UHV are given below.

4.1.1 Vane-Pump

The Rotary vane pump is used to control the gas flow inside the UHV from high pressure area to low pressure area. The vane is linked to a rotor and at the time of rotating, it inlets gas to in one side and the gas gets compressed in other side. Lastly the gas leaves the chamber through exhaust pipe shown in the figure.

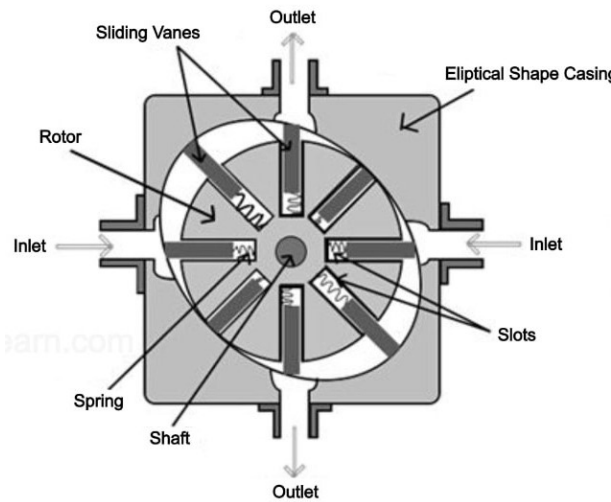


Figure 4: Schematic Diagram of Rotary Vane Pump [3]

4.1.2 Turbo-Molecular Pump

Another most important pump is turbo-molecular pump which helps to maintain low pressure about 10^{-10} . It consists of rotor, stationary stator blades, outlet and inlet path.

More and more gas gets compressed while the the blades rotates and after many times repetition,the air leaves the chamber.

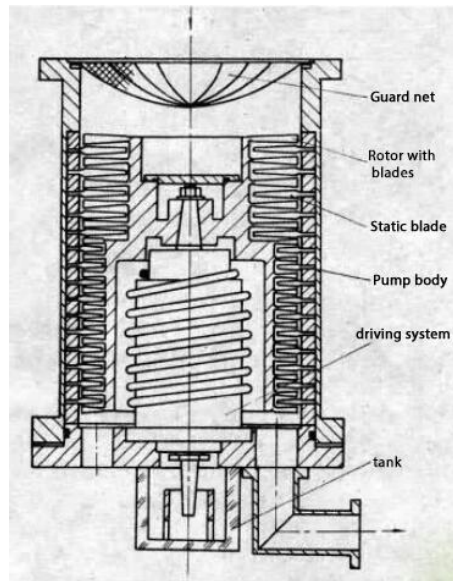


Figure 5: Schematic Diagram of Rotary Vane Pump[Ref-2]

4.1.3 Ion-Getter Pump

The ion-pupm helps to remove the gas through ionization process.It consists of anode and cathode that are installed inside a magnet.Since the inletted gas gets ionized,the gas is eventually removed by this process.

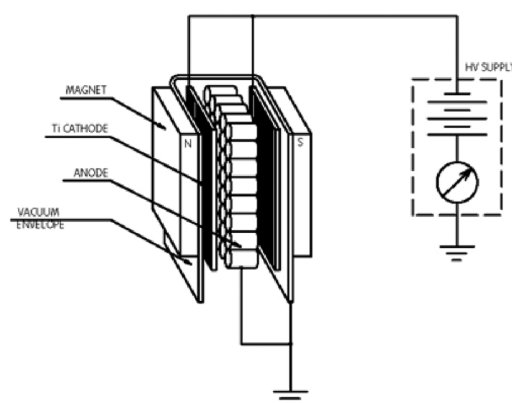


Figure 6: Schematic Diagram of Rotary Vane Pump [4]

4.2 Cylindrical Mirror Analyzer(CMA)

A cylindrical mirror analyzer integrated with a electron gun is installed inside the ultra-high vacuum chamber. The components of CMA are given below.

4.2.1 Electron gun

The electron gun is used to generate electrons which is then accelerated by high voltage

4.2.2 Two concentric cylinders

Two concentric cylindrical mirror is used to filter the electrons having certain kinetic energy. The entrance and exit slits only allow electrons to pass with certain angle, this way filtered out.

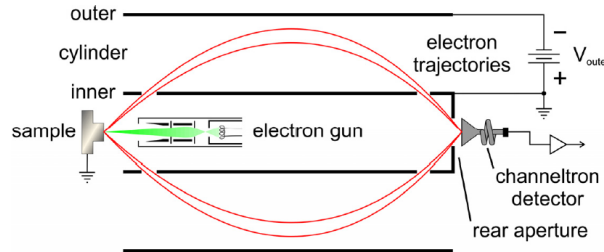


Figure 7: Schematic cross-section of cylindrical mirror analyzer(CMA) integrated with a electron gun [5]

4.2.3 Channeltron detector

It works based on secondary electron multiplier process and traces the electrons with narrow band coming from the concentric cylinder mirror

4.3 Lock-in Amplifier

The lock-in amplifier works based on demodulation process which enhance the signal quality through amplification. When a signal is too small buried in high noise, in that case separation of expected signal is quite difficult. It uses a phase sensitive detection process by multiplying the signal by a reference signal whose frequency is already known which is integrated over a long period of time. This process only allows to keep the signal as same as the reference frequency.

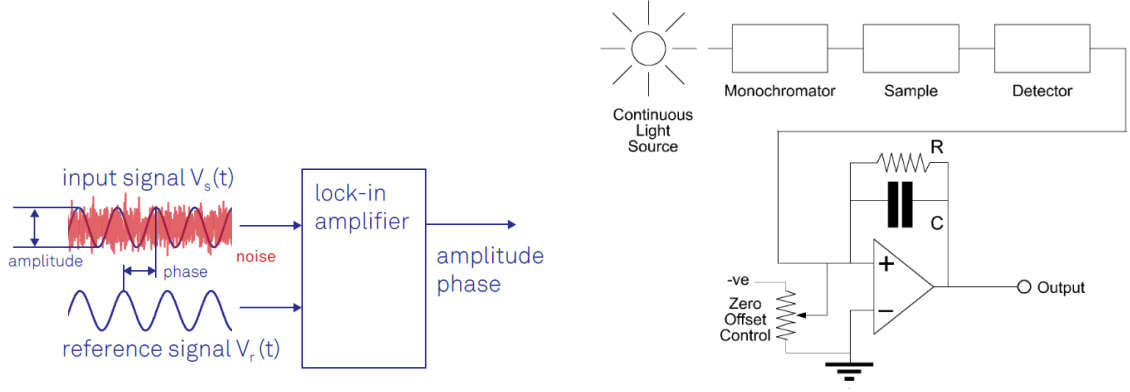


Figure 8: Lock-in amplifier [6]

5 Result and Analysis

In every obtained graph during experiment we plotted $\frac{dN(E)}{dE}$ with respect to given energy. That means we will get the derivative of the Gaussian peak in graph, implying that peaks can be find where the derivative corresponds to zero value.

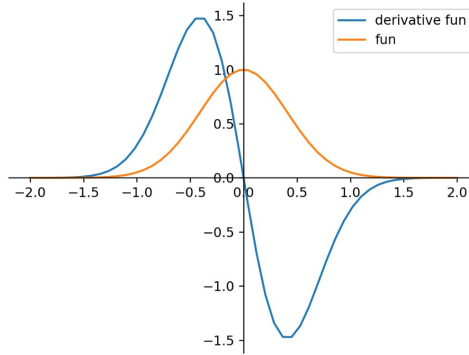


Figure 9: Gaussian and its derivative

5.1 Dirty Sample

taking a range from 0 to 1400eV we plotted the graph of $\frac{dN(E)}{dE}$ and we can observe the peaks at 72eV, 266eV, 516eV, 1231.4eV, 1393.5eV, after checking it from literature we can finally tell the impurity is O, C and Al respectively[7].

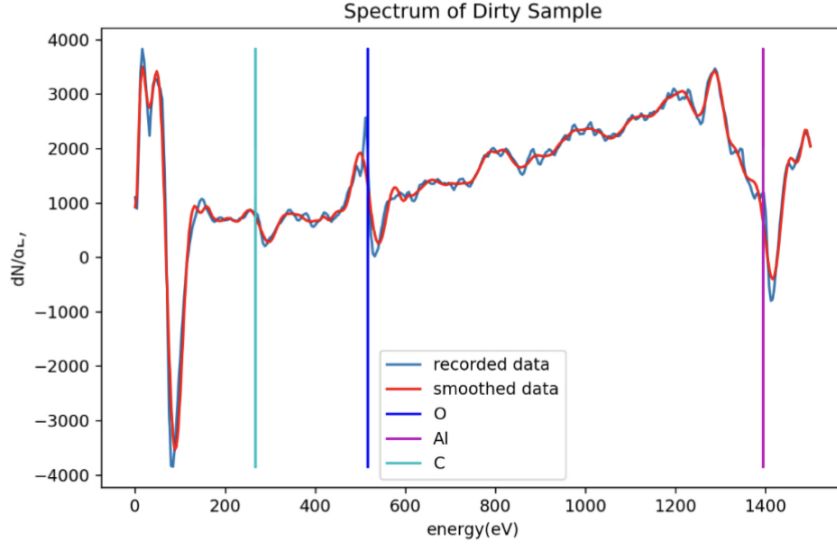


Figure 10: Spectrum of Dirty Sample

5.2 Cleaned Sample

In this figure we can observe that the peak corresponding to the energy 516eV is not there, that means the sample is clean now and we can move to the next tasks.

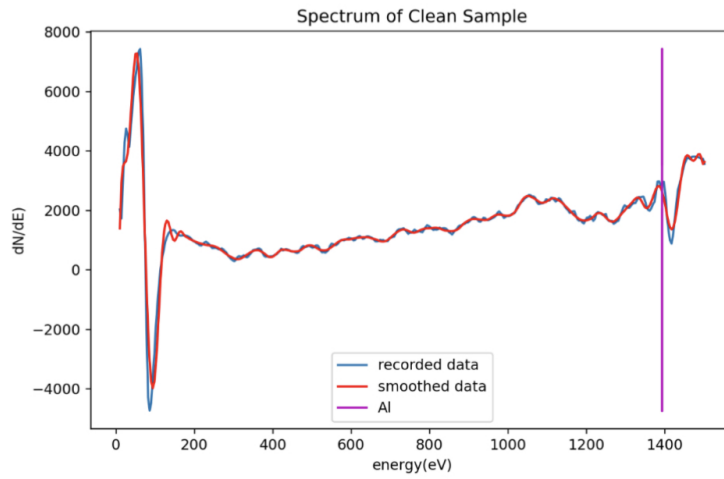


Figure 11: Spectrum of Clean Sample

5.3 Auger Spectrum of Aluminum

The energies of auger electrons is close to 1400. So, to get the spectrum of auger electrons, we set the range of the energy on graph between 1250eV-1450eV. Now to get the better signal from the sample we change three parameters phase, time constant and sensitivity. Phase matching that is by changing the phase of the reference signal will give us the optimized signal, time constant will give us the period of integration that will smoothen the signal and sensitivity will give us the better resolution. Now after checking it for a lot of combinations we got a better signal at phase 155 time constant 300ms and sensitivity $300\mu V$. We can see in the figure 12 our plot for auger spectrum. Here peaks define the different transitions of electrons involved in auger electron emission. The red curve is the smoothed curve which gives the approximately accurate values of the energies of the transition of electrons. The vertical lines define the derivative of different Gaussian peaks. So, we put these values of energies associated with different levels of transitions in the table below:

Auger Line	Theoretical Value (eV)	Observed Value (eV)
$KL_{2,3}L_{2,3}(^1D)$	1393.5	1393.5 ± 0.5
$KL_{2,3}L_{2,3}(^1D) + 1B.P$	1378.3	1380.3 ± 0.5
$KL_1L_{2,3}(^3P)$	1358.5	1360.7 ± 0.5
$KL_1L_{2,3}(^1P)$	1342.3	1341.5 ± 0.5
$KL_1L_{2,3}(^1P) + 1B.P$	1327.1	1324.9 ± 0.5
$KL_1L_1(^1S)$	1304.0	1305.5 ± 0.5

Table 1: Energy of Transition

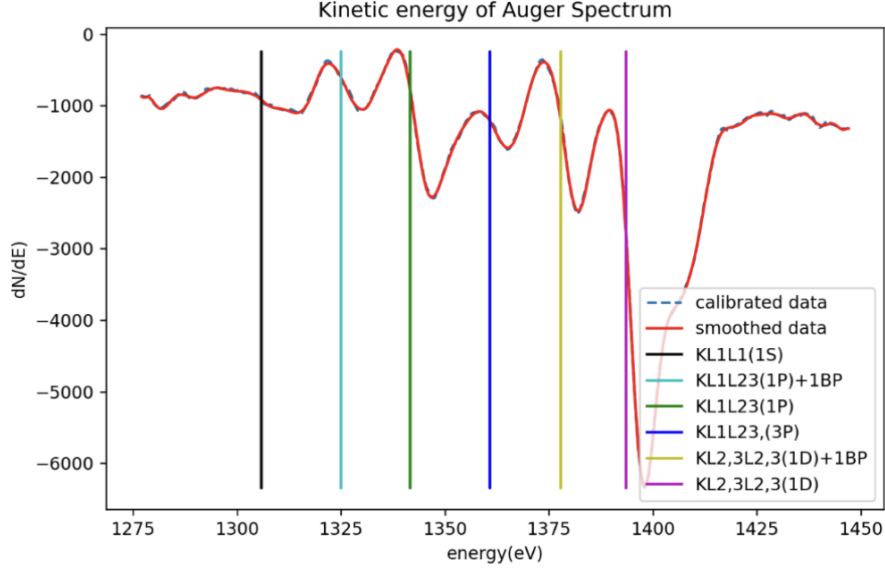


Figure 12: Auger Spectrum of Al

5.4 Electron Energy loss Spectrum for Bulk and Surface Plasmons

The principle about mean free path of an inelastic electron is the reason for the formation of bulk and surface plasmons. The energy associated with the plasmons can be calculated by calculating the difference in energies of elastically and in-elastically scattered electron. Depending upon the plasmon type, the incident energy should be in accordance to it. For bulk plasmon we need more energetic electrons than the incident electrons for formation of surface plasmons. This is because bulk plasmons are made deep in the material, so to penetrate it we need more energy.

5.5 Bulk Plasmon

To detect bulk plasmons we set our energy of electrons equal to 1000eV and the set the energy range from 920eV to 1070eV. After calculating spectrum for many combinations, the better one was found at phase 155, Sensitivity 300mV and the time constant is 100ms. The theoretical value of energy for bulk plasmon is 15.2eV. We calculate the energy of bulk plasmons by subtracting the energies of inelastic electrons in a table below from the elastic electron which is 1049.8eV. After comparing the energy values with the value from literature 15.2eV we can determine the number of bulk plasmons created.

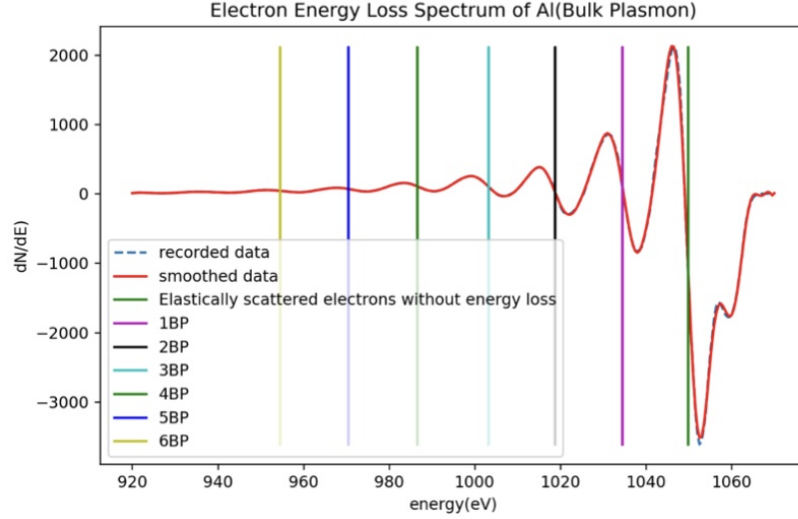


Figure 13: Determination of Bulk Plasmon using EEL spectrum of Al

Plasmon	Peak Position (eV)	Energy per Plasmon (eV)
1 B.P	1034.4	15.4 ± 0.1
2 B.P	1018.7	15.5 ± 0.1
3 B.P	1003.2	15.5 ± 0.1
4 B.P	986.5	15.8 ± 0.1
5 B.P	970.5	15.9 ± 0.1
6 B.P	954.5	15.9 ± 0.1

Table 2: Bulk Plasmons Energies EEL

5.6 Surface Plasmon

To detect surface plasmons we set our energy of electrons equal to 200eV and the set the energy range from 150eV to 280eV. After calculating spectrum for many combinations, the better one was found at phase 155, Sensitivity 1V and the time constant is 300ms. The theoretical value for the energy of a surface plasmon is 10.4eV We calculate the energy of surface plasmons by subtracting the energies of inelastic electrons in a table below from the elastic electron which is 208.35eV.

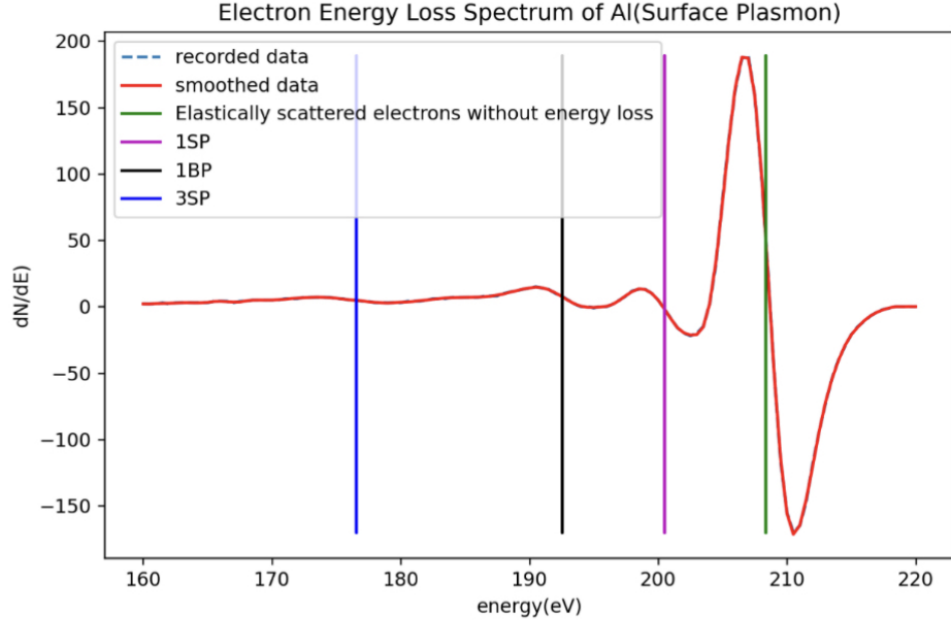


Figure 14: Determination of Surface Plasmon using EEL spectrum of Al

Plasmon	Peak Position (eV)	Energy of Plasmon (eV)
1 S.P	200.5	7.9 ± 0.1
2 S.P or 1B.P	192.51	$2(7.9 \pm 0.1)$ or 15.8 ± 0.1
3 S.P	176.5	$3(10.06 \pm 0.1)$

Table 3: Surface Plasmons Energies EEL

6 Conclusion

Auger electron spectroscopy can be used to get KLL spectrum of Al and other elements till transition metals, in addition it can also be used to detect the impurity present in material which then can be removed by different physical and chemical processes. The electron energy loss spectroscopy gives us the spectrum associated with quantum of quasiparticles called plasmons. Depending upon the energy lost by electrons through inelastic collision will define the type of plasmon that is bulk or surface plasmon.

References

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